

LEVEL 1: EASY (Square & Higher-Index Monomial Denominators)

Exercise 1

Rationalize the denominator:

$$\frac{8}{\sqrt{6}}$$

Answer 1

1. Multiply top and bottom by $\frac{\sqrt{6}}{\sqrt{6}}$:

$$\frac{8}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{8\sqrt{6}}{6}$$

2. Simplify the fraction by dividing by 2:

$$= \frac{4\sqrt{6}}{3}$$

Exercise 2

Rationalize the denominator:

$$\frac{9}{4\sqrt{3}}$$

Answer 2

1. Multiply numerator and denominator by $\frac{\sqrt{3}}{\sqrt{3}}$:

$$\frac{9}{4\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{3}}{4 \cdot 3} = \frac{9\sqrt{3}}{12}$$

2. Reduce by dividing numerator and denominator by 3:

$$= \frac{3\sqrt{3}}{4}$$

Exercise 3

Rationalize the denominator:

$$\frac{2}{\sqrt[3]{3}}$$

Answer 3

1. Multiply by $\frac{\sqrt[3]{3^2}}{\sqrt[3]{3^2}}$ to complete exponent $n = 3$:

$$\frac{2}{\sqrt[3]{3}} \cdot \frac{\sqrt[3]{9}}{\sqrt[3]{9}}$$

2. Combine terms in the denominator:

$$= \frac{2\sqrt[3]{9}}{\sqrt[3]{3^3}}$$

3. Simplify root:

$$= \frac{2\sqrt[3]{9}}{3}$$

LEVEL 2: INTERMEDIATE (Monomial Higher-Index & Basic Conjugates)

Exercise 4

Rationalize the denominator:

$$\frac{15}{\sqrt[4]{27}}$$

Answer 4

1. Express 27 as 3^3 , so multiplier is $\frac{\sqrt[4]{3^{4-3}}}{\sqrt[4]{3}} = \frac{\sqrt[4]{3}}{\sqrt[4]{3}}$:

$$\frac{15}{\sqrt[4]{3^3}} \cdot \frac{\sqrt[4]{3}}{\sqrt[4]{3}} = \frac{15\sqrt[4]{3}}{\sqrt[4]{3^4}} = \frac{15\sqrt[4]{3}}{3}$$

2. Simplify fraction:

$$= 5\sqrt[4]{3}$$

Exercise 5

Rationalize the denominator:

$$\frac{12}{\sqrt{11} - \sqrt{5}}$$

Answer 5

1. Multiply by conjugate $(\sqrt{11} + \sqrt{5})$:

$$\frac{12}{\sqrt{11} - \sqrt{5}} \cdot \frac{\sqrt{11} + \sqrt{5}}{\sqrt{11} + \sqrt{5}}$$

2. Apply difference of squares $(a-b)(a+b) = a^2 - b^2$:

$$= \frac{12(\sqrt{11} + \sqrt{5})}{11 - 5} = \frac{12(\sqrt{11} + \sqrt{5})}{6}$$

3. Cancel factor 6:

$$= 2\sqrt{11} + 2\sqrt{5}$$

Exercise 6

Rationalize the denominator:

$$\frac{10}{3 - \sqrt{2}}$$

Answer 6

1. Multiply by conjugate $(3 + \sqrt{2})$:

$$\frac{10}{3 - \sqrt{2}} \cdot \frac{3 + \sqrt{2}}{3 + \sqrt{2}}$$

2. Apply difference of squares in denominator:

$$= \frac{10(3 + \sqrt{2})}{3^2 - (\sqrt{2})^2} = \frac{30 + 10\sqrt{2}}{9 - 2}$$

3. Simplify:

$$= \frac{30 + 10\sqrt{2}}{7}$$

Exercise 7

Rationalize the denominator:

$$\frac{\sqrt{5}}{\sqrt{5} + \sqrt{2}}$$

Answer 7

1. Multiply top and bottom by conjugate ($\sqrt{5} - \sqrt{2}$):

$$\frac{\sqrt{5}}{\sqrt{5} + \sqrt{2}} \cdot \frac{\sqrt{5} - \sqrt{2}}{\sqrt{5} - \sqrt{2}}$$

2. Distribute numerator and simplify denominator:

$$= \frac{(\sqrt{5})^2 - \sqrt{10}}{5 - 2} = \frac{5 - \sqrt{10}}{3}$$

$$= \frac{5 - \sqrt{10}}{3}$$

LEVEL 3: HARD (Complex Radicals, Multi-Step & Algebraic Binomials)

Exercise 8

Rationalize the denominator and simplify fully:

$$\frac{\sqrt{10}}{\sqrt{5} - \sqrt{3}}$$

Answer 8

1. Multiply by conjugate ($\sqrt{5} + \sqrt{3}$):

$$\frac{\sqrt{10}}{\sqrt{5} - \sqrt{3}} \cdot \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} + \sqrt{3}} = \frac{\sqrt{50} + \sqrt{30}}{5 - 3}$$

2. Simplify radical $\sqrt{50} = 5\sqrt{2}$:

$$= \frac{5\sqrt{2} + \sqrt{30}}{2}$$

$$= \frac{5\sqrt{2} + \sqrt{30}}{2}$$

Exercise 9

Rationalize the denominator and simplify radicals:

$$\frac{18}{\sqrt[5]{8}}$$

Answer 9

1. Write $8 = 2^3$, then multiplier is $\frac{\sqrt[5]{2^{5-3}}}{\sqrt[5]{2^2}} = \frac{\sqrt[5]{2^2}}{\sqrt[5]{2^2}}$:

$$\frac{18}{\sqrt[5]{2^3}} \cdot \frac{\sqrt[5]{4}}{\sqrt[5]{4}} = \frac{18\sqrt[5]{4}}{\sqrt[5]{2^5}} = \frac{18\sqrt[5]{4}}{2}$$

2. Simplify:

$$= 9\sqrt[5]{4}$$

Exercise 10

Combine into a single fraction and simplify:

$$\frac{3}{\sqrt{7}-2} - \frac{3}{\sqrt{7}+2}$$

Answer 10

1. Rationalize each term using its conjugate:

$$\frac{3(\sqrt{7}+2)}{7-4} = \frac{3(\sqrt{7}+2)}{3} = \sqrt{7}+2$$

$$\frac{3(\sqrt{7}-2)}{7-4} = \frac{3(\sqrt{7}-2)}{3} = \sqrt{7}-2$$

2. Subtract terms:

$$(\sqrt{7}+2) - (\sqrt{7}-2) = \sqrt{7}+2 - \sqrt{7}+2$$

$$= 4$$

Exercise 11

Simplify fully for $x > 0, x \neq 9$:

$$\frac{x-9}{\sqrt{x}+3}$$

Answer 11

1. Multiply by the conjugate $(\sqrt{x}-3)$:

$$\frac{x-9}{\sqrt{x}+3} \cdot \frac{\sqrt{x}-3}{\sqrt{x}-3}$$

2. Apply difference of squares to denominator:

$$= \frac{(x-9)(\sqrt{x}-3)}{(\sqrt{x})^2 - 3^2} = \frac{(x-9)(\sqrt{x}-3)}{x-9}$$

3. Cancel factor $(x-9)$:

$$= \sqrt{x}-3$$